

Aluminum

The invisible deficit is about to become visible

- Even with higher Chinese supply forecasts and a base case that Middle East smelters begin the process of ramping up idled production next quarter, we still forecast a more than 2 mmt aluminum deficit over 2Q26-4Q26.
- For now, this supply gap is being bridged first by a draw in “invisible” inventories (producer/trader/consumer stocks).
- Yet these hidden stocks are also now depleting, with industry feedback suggesting only ~2 more months of drawdown coverage.
- With our balances showing a primary aluminum deficit of nearly 1 mmt that still needs to be covered in 2H26, the deficit should increasingly transmit into draws in visible inventory, which are heavily concentrated (75%) in China.
- A sustained, material drawing trend in Chinese inventory should support higher SHFE prices. To keep needed Chinese product exports flowing, LME prices must stay high enough to keep the export arb open. This arb chase is the next core fundamental driver for higher LME prices.
- While aluminum could come under initial knee-jerk pressure following a sustained re-opening of the Strait of Hormuz, we don’t think it will be sustained as Middle East production will still likely take multiple quarters to approach normalcy, keeping deficits in place.
- In short, the sharp deficit in aluminum is taking time to transmit through the supply chain, a process that will continue even after the Strait reopens. We are in the middle innings and still think aluminum prices are biased higher in the coming quarters.
- We still expect an ultimate push towards \$4,000/mt and now see aluminum prices averaging \$3,750/mt over 2H26.
- Chinese aluminum export controls, motivated by ensuring domestic supply security, remain a significant bullish risk in our view.
- Accelerating Chinese-funded supply outside China (Indonesia, Angola, Saudi, etc.) does little to change the supply gap in the near term, but longer term it is a momentous shift, essentially ending a period of global supply discipline in aluminum.
- Even as overall supply and demand remains relatively balanced in 2027, we think prices will begin to roll over next year, falling back below \$3,000/mt in 2H27 as the spectre of looming 2028 oversupply, which is now approaching 1 mmt, comes into sharper focus.

Global Commodities Research

Gregory C. Shearer

(44-20) 7134-8161
gregory.c.shearer@jpmorgan.com
J.P. Morgan Securities plc

Ali A. Ibrahim

(44-20) 3493-6438
ali.ibrahim@jpmorgan.com
J.P. Morgan Securities plc

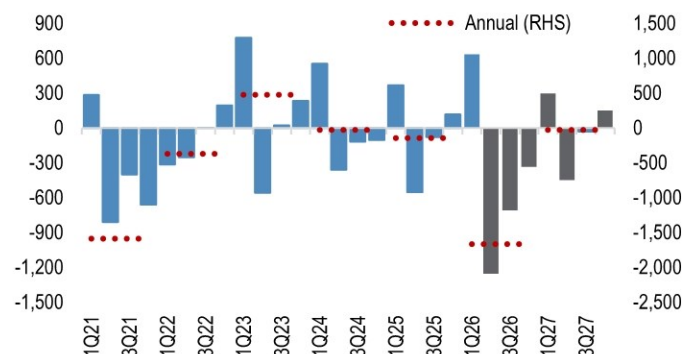
Ananyashree Gupta

(91-22) 6157 3627
ananyashree.gupta@jpmchase.com
J.P. Morgan India Private Limited

The invisible to become visible. Despite boosting estimates for Chinese supply even further and continuing to assume a more imminent opening of the Strait of Hormuz that allows Middle East smelters to begin the process of ramping up idled production next quarter, in our latest aluminum balance update we still forecast a monumental supply hole that needs to be filled. Overall, we now forecast a 1.7 mmt primary aluminum deficit in 2026 with 2Q26-4Q26 amounting to an implied 2.3 mmt draw on inventory over three quarters (**Figure 1**). With our assumption that Middle East smelter restarts begin in 3Q26 and accelerate over 4Q26 and 1Q27, this current quarter marks the peak deficit in our forecast with a 1.3 mmt implied draw in 2Q26 (**Figure 2**). However, while the push higher in global premiums reflects this tightness, global visible aluminum inventory, after drawing in the last few weeks, is basically still only roughly on par with end-February levels (**Figure 3 & Figure 4**).

Figure 1: Global primary aluminum balance, quarterly

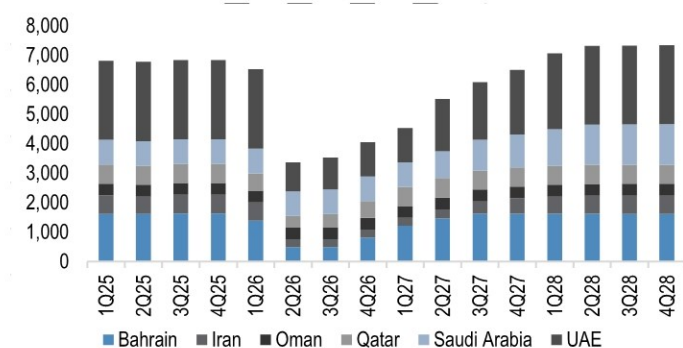
Thousand mt



Source: Company Reports, Government and Industry data, IAI, CRU, J.P. Morgan Commodities Research

Figure 2: JPM estimates for Middle East primary aluminum production, quarterly

Thousand mt, annualized

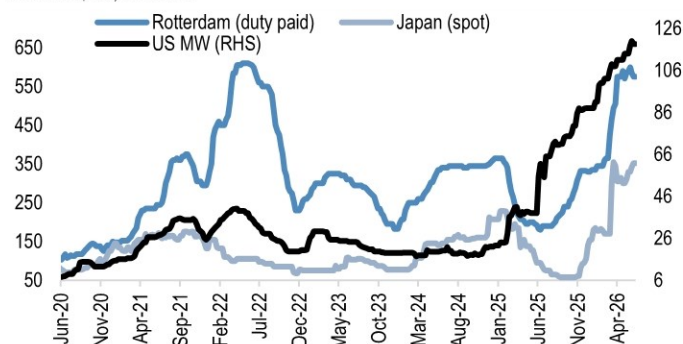


Source: Company reports, CRU, J.P. Morgan Commodities Research

In our view this mismatch points to a significant, industry-wide destock of more “invisible” inventory, stocks of aluminum not captured in exchange and social inventory tallies, being the first mechanism called upon to backfill the chasm in supply this quarter. This includes stocks of metal held by Middle East producers in consumer regions (Asia, Europe, US) which have helped buffer the blow from disrupted production and logistics so far. Moreover, this also includes trader and merchant stocks not held at the exchange to avoid the higher LME rent. Given the backwardated forward curve for most of this quarter, which penalizes stock holders who need to finance and carry this metal, there has been strong incentive to sell off this inventory too and for consumers themselves to draw down their carrying inventory buffer as well to avoid the backwardation penalty as much as possible.

Figure 3: Regional aluminum ingot premiums

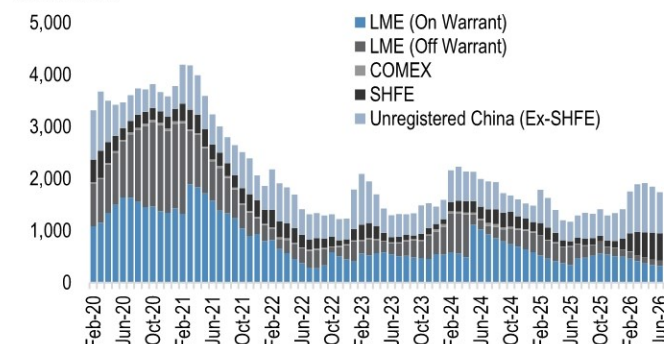
LHS: US\$/mt; RHS: c/lb



Source: CRU

Figure 4: Global visible aluminum inventories

Thousand mt



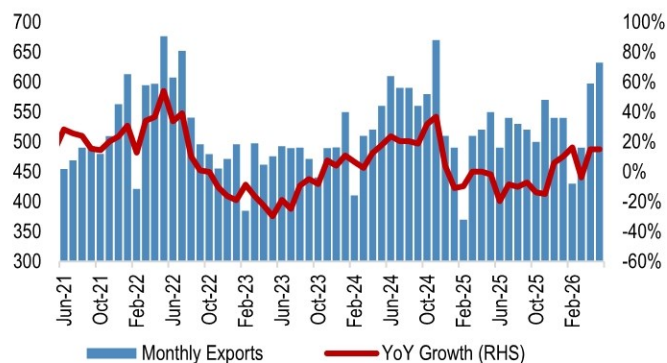
Source: CRU, LME, SHFE, COMEX, SMM. Regional includes stocks in Shanghai, Jiangsu, Nanhai, Zhejiang, Tianjin, Henan, Shandong, Chongqing.

Feedback from our discussions around recent industry conferences, including the Harbor Aluminum Summit last week, indicate that these hidden pools of inventory are now also running lean, with general expectations that there is now around 2 more months of drawdown coverage of this more “invisible” inventory. However, our balance still shows a primary aluminum deficit of nearly 1 mmt that still needs to be covered in 2H26 (with a 700 kmt deficit in 3Q26) as, even when the Strait of Hormuz more fully reopens (our base case is this month), smelter ramp ups take time, even 1-2 quarters following a controlled shutdown. As invisible industry stockpiles begin to exhaust in the coming months, we think the ongoing aluminum deficit will begin to become much more “visible”, necessitating draws from global surveyed stocks where China holds about 75% of the global ~1.7 mmt of visible inventory.

In short, the sharp deficit in aluminum is taking time to transmit through the supply chain, a process that will continue even after the Strait reopens. We are in the middle innings and still think aluminum prices are biased higher in the coming quarters as the next stretch involves a transition towards sharper draws in visible inventory concentrated in China. Given the magnitude of the supply shortage facing the market, the ex-China market needs China to continue to backfill for lost Middle East tonnes via boosted levels of aluminum product exports (**Figure 5**). While a jump higher in Chinese production to push above the 45 mmt capacity cap amid expanded producer margins combined with softer domestic ingot demand for now have both helped so far to insulate China’s inventory level from significant draws, we don’t think this can sustain. Indeed, this week brought a ~63 kt wow draw in Chinese social inventory, the largest weekly draw since December 2023 (**Figure 6**). Moreover, while we see Chinese supply pushing above 45 mmt on maximized efficiency gains at some smelters, we do think ongoing inspections and greater scrutiny of key industrial energy use and emissions will likely act as a brake, preventing supply from significantly blowing past the capacity cap (**Figure 7**).

Figure 5: Chinese unwrought aluminum and aluminum product exports

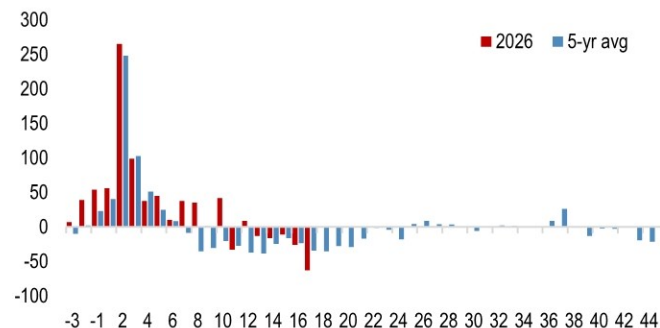
LHS: Thousand mt; RHS: Percent change, yoy



Source: China Customs, J.P. Morgan Commodities Research

Figure 6: Weekly change in China visible aluminum inventory (SHFE + regional warehouses).

X-axis = weeks post Chinese New Year (week 0 = week closest to start of CNY), Y-axis = kt of aluminum increase / (decrease)

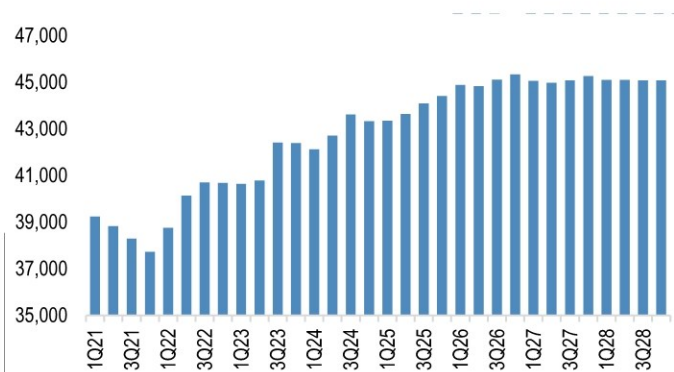


Source: SHFE, CRU, SMM, J.P. Morgan Commodities, *First and second weeks grouped together due to lagged inventory reporting.

Eventually, given we still see 1.6% yoy domestic demand growth in China this year (2.5% yoy growth inclusive of boosted exports), the continued excess export pull on Chinese supply from the rest of the world is expected to drive a more pronounced and prolonged visible destocking in Chinese inventory over the coming months (**Figure 8**). Amid this sharper destocking and relative tightening in China's inventory coverage, we see Chinese SHFE aluminum prices pushing higher. With transaction prices in the rest of the world needing to keep China's product export arb open to receive sufficient Chinese metal, we see this arb friction being the fundamental catalyst for LME aluminum prices to continue to push higher (**Figure 10**).

Figure 7: Annualized Chinese quarterly primary aluminum production estimates

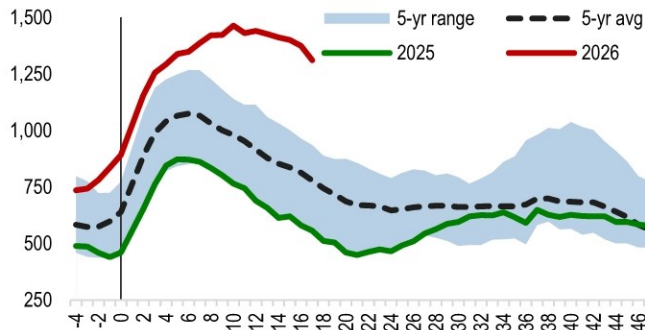
Thousand mt



Source: CRU, J.P. Morgan Commodities Research

Figure 8: Total visible China aluminum inventory (SHFE + regional warehouses)

Y-axis: Thousand mt; X-axis: Weeks around Chinese New Year (0 = week closest to start of CNY)



Source: SHFE, CRU, SMM, J.P. Morgan Commodities Research

The journey is taking longer but the destination remains the same. With our updated balance views, we remain bullish on aluminum over 2H26 and still see LME prices pushing towards \$4,000/mt on a spot basis in the near term. So far this week, aluminum has come under pressure amid a broader risk-off move, falling back towards \$3,500/mt and consumer buying support as the backwardation at the front of the curve eased significantly. We don't think this bearish bout of volatility is a sea change, rather, we

view it as another breather in aluminum's eventual push higher as the underlying fundamentals in the coming quarters remain broadly the same... very tight. Moreover, while aluminum could come under initial knee-jerk pressure with energy prices following a sustained re-opening of the Strait of Hormuz, we don't think it will be sustained. With supply loss still solidified for months amid slow Middle East restarts (a reopening by mid-year is indeed our base case) and risks to overall demand and the macroeconomy actually diminished, we see this outcome ultimately skewing bullish amid a continued deficit market. With the journey towards visible inventory draws taking longer than initially expected, but our conviction in a very large supply shortfall still high, we still expect an ultimate push towards \$4,000/mt occurring in the coming quarters, with aluminum averaging \$3,750/mt over 2H26 (**Figure 9**).

Figure 9: JPM aluminium price forecasts

LME cash in US\$ per metric tonne, quarterly and annual averages

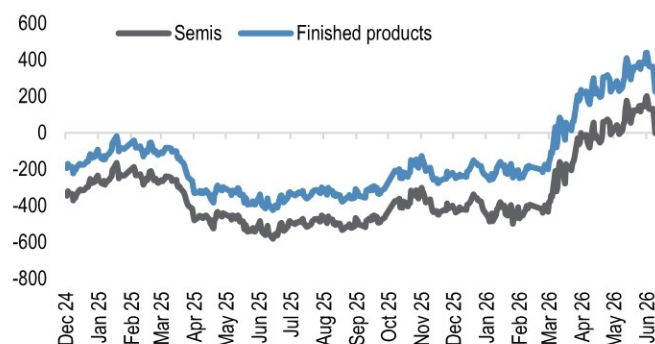
		4Q2025A	2025A	1Q2026A	2Q2026	3Q2026	4Q2026	2026	1Q2027	2Q2027	3Q2027	4Q2027	2027
Aluminum	New	2,828	2,631	3,198	3,650	3,800	3,700	3,587	3,450	3,200	2,900	2,750	3,075
	Old (Apr 2026)	2,828	2,631	3,198	3,800	3,600	3,400	3,500	3,200	3,000	2,900	2,750	2,963
	Change	0%	0%	0%	-4%	6%	9%	3%	8%	7%	0%	0%	4%

Source: J.P. Morgan Commodities Research

No clear evidence of demand destruction. Stronger-than-expected Chinese supply growth and quicker Middle East restarts/ramp-ups remain the most credible bearish fundamental risks to our view. Demand destruction at these higher prices and amid a broader inflationary environment also remains a risk but one that, for now, isn't showing immediate blaring warning signs. In fact, at our meetings last week at Harbor, the mood and commentary on demand was outright upbeat and we really didn't encounter any consumers who reported evidence or serious concerns of demand destruction amid higher prices. While this admittedly is a US-centric view and potentially influenced by a weaker demand base last year, at the same time, the US consumer also is currently paying nearly \$2,000/mt more for every tonne of aluminum consumed now given tariffs. Hence, if there was a clear, distilled read on the threat of price-induced demand destruction given the fresh rally, one would think it would first most acutely show up in the US market. All this is not to say that we think global aluminum demand is going to be spectacular over the balance of the year, but we think it will be OK. We still see around +0.4% yoy ex-China demand growth, with European demand quite subdued, summing to around +1.7% growth yoy globally when adding in China's export-boosted 2.5% yoy primary demand growth this year (**Figure 11**).

Figure 10: China aluminum semis & products export arb proxy (China to rest of Asia)

US\$/mt. Differential between China spot prices (inclusive of VAT) and LME + Asia premium

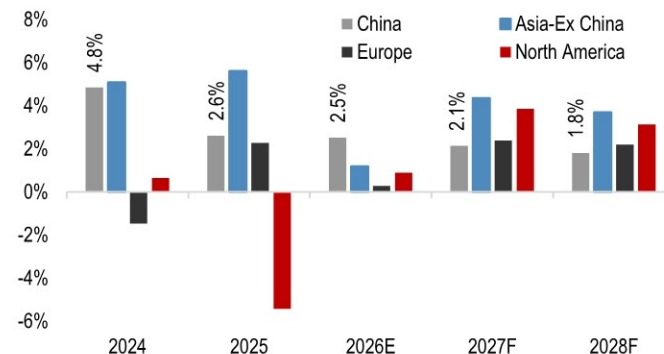


Note: Export arb is calculated as the difference between China spot prices and LME + Asia premium less conversion costs (ingot to semis & semis to finished products), logistics, shipping and financing costs.

Source: SHFE, SMM, Shanghai Metals Exchange, LME, Bloomberg L.P., J.P. Morgan Commodities Research

Figure 11: Primary aluminium demand growth forecasts by region

Percent change yoy



Source: CRU, J. P. Morgan Commodities Research

Chinese export controls would be a significantly bullish twist. While China has comfortable aluminum stock coverage for now, if our fundamental view eventuates, Chinese inventory will begin to more materially draw over the coming months, primarily due to significantly boosted exports to the rest of the world. If Chinese stocks are lower and drawing sharply, we do think there is the risk (albeit not our base case at this time) that Chinese policymakers become more concerned about aluminum supply security, particularly if Guinea, which accounts for around three-quarters of China's bauxite imports, follows through with more [material curbs on bauxite exports](#), as planned from this month forward. With potentially growing anxiety that China is feeding the rest of world's aluminum shortage at the expense of domestic security of supply, Chinese policymakers could begin to move to curtail aluminum exports through either enacting additional export tariffs on aluminum products and/or outright limiting or restricting exports. If this risk were to come to fruition, LME aluminum prices would likely have significant upside even above our bullish price forecast over the balance of 2026. If an export tariff comes, LME prices likely need to price it in to keep incentivizing product exports, whereas outright restrictions would likely quickly exhaust LME inventory as LME prices rise to destroy demand and solve for the supply shortage in the coming quarters without the help of Chinese tonnes.

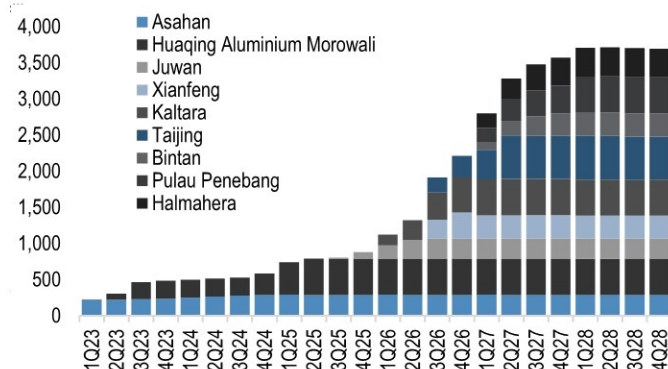
Get ready for the whiplash: The near-term supply hole in aluminum is very large but it also is temporary... and 2028 balances are starting to look very sloppy. While the ultimate timing of Middle East restarts remains fluid (and a longer-than-expected closure of the Strait risks prolonging the current period of tightness), ultimately when these smelters fully ramp up again, they will be delivering into a market where Chinese-funded supply outside of China has sharply increased. As has been the trend in every recent balance update, our Indonesia supply numbers over the forecast horizon have once again risen materially on the back of both quicker ramp-ups and greater overall production capacity coming online. By 4Q27, our current estimate for the annualized production rate in Indonesia is about 2.5 mmt higher than the production level in 1Q26 (**Figure 12**). And it is now not only Indonesia. Chinese-funded projects in Angola (Barra do Dande, Tianyuan Angola) and Saudi Arabia (Red Sea Aluminum) are being accelerated in our base case too, while announcements of (large) project intentions have

sprung up in countries ranging from Kazakhstan to Oman. We still think this does little to change the supply gap in the near term, but longer term it is a momentous shift.

Essentially, the era of global aluminum supply discipline given the 45 mmt cap in China is ending and with its demise our current 2028 surplus has grown to now approach 1 mmt even with some pretty punchy demand numbers embedded for now. This means that on the other end of the current supply deficit, which has significantly boosted producer margins, we see a growing surplus emerging which will once again pressure prices back towards costs, eroding these bumper producer margins in a bid to stem the gusher of new supply. Even as overall supply and demand remains relatively balanced in 2027, we think this price rollover likely begins in earnest next year, with prices falling below \$3,000/mt in 2H27 as the spectre of looming 2028 oversupply draws into sharper focus (**Figure 13**).

Figure 12: Indonesian primary aluminium production by smelter, quarterly

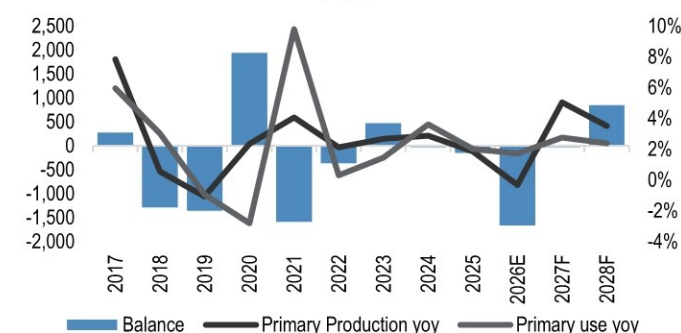
Thousand mt, annualised



Source: CRU, various news and company reports, J.P. Morgan Commodities Research

Figure 13: Global primary aluminum balance and supply and demand growth

LHS: Thousand mt; RHS: Percent change, yoy



Source: Company Reports, Government and Industry data, IAI, CRU, J.P. Morgan Commodities Research

Figure 14: Primary aluminium supply and demand balance

Thousand metric tonnes

	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027F	2028F
Primary Production	63,612	63,961	63,278	64,765	67,398	68,834	70,681	72,721	74,070	73,830	77,557	80,283
growth	7.8%	0.5%	-1.1%	2.4%	4.1%	2.1%	2.7%	2.9%	1.9%	-0.3%	5.0%	3.5%
Primary Use	63,334	65,246	64,636	62,822	68,984	69,202	70,205	72,750	74,216	75,495	77,584	79,435
growth	6.0%	3.0%	-0.9%	-2.8%	9.8%	0.3%	1.4%	3.6%	2.0%	1.7%	2.8%	2.4%
Balance	278	-1,285	-1,358	1,944	-1,586	-368	476	-29	-146	-1,665	-28	848

Source: Company Reports, Government and Industry data, IAI, CRU, Wood Mackenzie, J.P. Morgan Commodities Research

Figure 15: Primary aluminium supply and demand balance by region

Thousand metric tonnes

Primary Production	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027F	2028F
Asia	40,627	41,046	40,038	41,337	43,724	45,469	47,183	48,901	50,035	52,122	54,235	54,989
China	36,374	36,326	35,359	36,753	38,523	40,079	41,573	43,072	43,879	45,047	45,101	45,222
North America	3,960	3,797	3,950	4,094	4,045	3,910	4,068	3,990	3,945	3,949	4,061	4,068
Central & South America	1,390	1,165	1,093	990	1,149	1,313	1,572	1,680	1,738	1,803	1,859	1,872
Europe	4,485	4,499	4,375	4,246	4,292	3,568	3,206	3,290	3,465	3,722	3,799	3,800
Eurasia	4,138	4,178	4,305	4,328	4,279	4,371	4,396	4,563	4,503	4,700	4,707	4,728
Middle East	5,482	5,670	5,958	6,243	6,423	6,706	6,734	6,809	6,822	4,286	5,573	7,227
Africa	1,704	1,689	1,643	1,609	1,595	1,652	1,634	1,624	1,683	1,332	1,370	1,643
Oceania	1,826	1,918	1,916	1,918	1,891	1,845	1,888	1,864	1,879	1,916	1,953	1,956
Primary Production	63,612	63,961	63,278	64,765	67,398	68,834	70,681	72,721	74,070	73,830	77,557	80,283
Primary Use	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027F	2028F
Asia	43,011	44,531	44,377	44,767	48,155	48,756	50,955	53,446	55,096	56,366	57,771	58,991
China	34,540	35,934	36,276	37,722	40,145	40,808	42,887	44,967	46,141	47,304	48,313	49,183
North America	6,587	6,692	6,388	5,474	6,211	6,215	5,854	5,892	5,572	5,622	5,839	6,022
Central & South America	1,209	1,156	1,129	1,016	1,217	1,174	1,144	1,235	1,312	1,342	1,398	1,432
Europe	9,218	9,397	9,267	8,356	9,787	9,611	8,877	8,747	8,946	8,972	9,187	9,389
Eurasia	896	946	931	909	983	801	833	875	797	817	839	866
Middle East	1,557	1,656	1,699	1,575	1,786	1,792	1,700	1,737	1,676	1,545	1,689	1,843
Africa	622	646	640	525	632	633	638	614	623	640	667	695
Oceania	234	223	205	200	213	220	204	204	194	192	195	197
Primary Use	63,334	65,246	64,636	62,822	68,984	69,202	70,205	72,750	74,216	75,495	77,584	79,435
Global Balance	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027F	2028F
Primary Production	63,612	63,961	63,278	64,765	67,398	68,834	70,681	72,721	74,070	73,830	77,557	80,283
Primary Use	63,334	65,246	64,636	62,822	68,984	69,202	70,205	72,750	74,216	75,495	77,584	79,435
Balance	278	-1,285	-1,358	1,944	-1,586	-368	476	-29	-146	-1,665	-28	848

Source: Company Reports, Government and Industry data, IAI, CRU, Wood Mackenzie, J.P. Morgan Commodities Research

JPM metals price forecasts

Quarterly and annual averages

	4Q25A	1Q26A	2Q26	3Q26	4Q26	1Q27	2Q27	3Q27	4Q27	2025	2026	2027
Base Metals												
Aluminum	2,828	3,198	3,650	3,800	3,700	3,450	3,200	2,900	2,750	2,631	3,587	3,075
Copper	11,105	12,824	13,500	13,000	12,500	11,800	11,600	11,600	11,500	9,947	12,956	11,625
Nickel	14,892	17,338	17,500	17,000	17,250	17,500	18,000	18,000	17,500	15,164	17,272	17,750
Zinc	3,165	3,237	3,420	3,500	3,400	3,300	3,200	3,050	3,000	2,867	3,389	3,138
Precious Metals												
Gold	4,152	4,873	4,800	5,300	6,000	6,200	6,250	6,300	6,300	3,440	5,243	6,263
Silver	55.2	83.7	78.5	85.0	90.0	88.0	87.0	85.0	83.0	40.1	84.3	85.8
Platinum	1,694	2,204	2,050	2,250	2,400	2,450	2,400	2,300	2,250	1,282	2,226	2,350
Palladium	1,478	1,706	1,490	1,550	1,600	1,550	1,500	1,450	1,400	1,151	1,586	1,475

Source: J.P. Morgan Commodities Research. *Base metals are LME cash and precious metals are spot.

Disclosures

History of Investment Recommendations:

A history of J.P. Morgan investment recommendations disseminated during the preceding 12 months can be accessed on the Research & Commentary page of <http://www.jpmorganmarkets.com> where you can also search by analyst name, sector or financial instrument.

Analysts' Compensation: The research analysts responsible for the preparation of this report receive compensation based upon various factors, including the quality and accuracy of research, client feedback, competitive factors, and overall firm revenues.

Company-Specific Disclosures: Important disclosures, including price charts and credit opinion history tables (if applicable), are available for compendium reports and all J.P. Morgan-covered companies, and certain non-covered companies, by visiting <https://www.jpmm.com/research/disclosures>, calling 1-800-477-0406, or e-mailing research.disclosure.inquiries@jpmorgan.com with your request.

Other Disclosures

J.P. Morgan is a marketing name for investment banking businesses of JPMorgan Chase & Co. and its subsidiaries and affiliates worldwide.

UK MIFID FICC research unbundling exemption: UK clients should refer to [UK MIFID Research Unbundling exemption](#) for details of J.P. Morgan's implementation of the FICC research exemption and guidance on relevant FICC research categorisation.

J.P. Morgan Research may, from time to time, write on issuers or securities targeted by economic or financial sanctions imposed or administered by the governmental authorities of the U.S., EU, UK or other relevant jurisdictions (Sanctioned Securities). Nothing in this report is intended to be read or construed as encouraging, facilitating, promoting or otherwise approving investment or dealing in such Sanctioned Securities. Clients should be aware of their own legal and compliance obligations when making investment decisions.

Any digital or crypto assets discussed in this research report are subject to a rapidly changing regulatory landscape. For relevant regulatory advisories on crypto assets, including bitcoin and ether, please see <https://www.jpmorgan.com/disclosures/cryptoasset-disclosure>.

The author(s) of this research report may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so.

Exchange-Traded Funds (ETFs): J.P. Morgan Securities LLC ("JPMS") acts as authorized participant for substantially all U.S.-listed ETFs. To the extent that any ETFs are mentioned in this report, JPMS may earn commissions and transaction-based compensation in connection with the distribution of those ETF shares and may earn fees for performing other trade-related services, such as securities lending to short sellers of the ETF shares. JPMS may also perform services for the ETFs themselves, including acting as a broker or dealer to the ETFs. In addition, affiliates of JPMS may perform services for the ETFs, including trust, custodial, administration, lending, index calculation and/or maintenance and other services.

Options and Futures related research: If the information contained herein regards options- or futures-related research, such information is available only to persons who have received the proper options or futures risk disclosure documents. Please contact your J.P. Morgan Representative or visit <https://www.theocc.com/components/docs/riskstoc.pdf> for a copy of the Option Clearing Corporation's Characteristics and Risks of Standardized Options or https://www.finra.org/sites/default/files/2020-08/Security_Futures_Risk_Disclosure_Statement_2020.pdf for a copy of the Security Futures Risk Disclosure Statement.

Changes to Interbank Offered Rates (IBORs) and other benchmark rates: Certain interest rate benchmarks are, or may in the future become, subject to ongoing international, national and other regulatory guidance, reform and proposals for reform. For more information, please consult: https://www.jpmorgan.com/global/disclosures/interbank_offered_rates

Private Bank Clients: Where you are receiving research as a client of the private banking businesses offered by JPMorgan Chase & Co. and its subsidiaries ("J.P. Morgan Private Bank"), research is provided to you by J.P. Morgan Private Bank and not by any other division of J.P. Morgan, including, but not limited to, the J.P. Morgan Corporate and Investment Bank and its Global Research division.

Legal entity responsible for the production and distribution of research: The legal entity identified below the name of the Reg AC Research Analyst who authored this material is the legal entity responsible for the production of this research. Where multiple Reg AC Research Analysts authored this material with different legal entities identified below their names, these legal entities are jointly responsible for the production of this research. Where more than one legal entity is listed under an analyst's name, the first legal entity is responsible for the production unless stated otherwise. Research Analysts from various J.P. Morgan affiliates may have contributed to the production of this material but may not be licensed to carry out regulated activities in your jurisdiction (and do not hold themselves out as being able to do so). Unless otherwise stated below in the legal entity disclosures, this material has been distributed by the legal entity responsible for production, or where more than one legal entity is listed under the analyst's name, the first legal entity will be responsible for distribution. If you have any queries, please contact the relevant Research Analyst in your jurisdiction or the entity in your jurisdiction that has distributed this research material.

Legal Entities Disclosures and Country-/Region-Specific Disclosures:

Argentina: JPMorgan Chase Bank N.A Sucursal Buenos Aires is regulated by Banco Central de la República Argentina ("BCRA"- Central Bank of Argentina) and Comisión Nacional de Valores ("CNV"- Argentinian Securities Commission - ALYC y AN Integral N°51).

Australia: J.P. Morgan Securities Australia Limited ("JPMSAL") (ABN 61 003 245 234/AFS Licence No: 238066) is regulated by the Australian Securities and Investments Commission and is a Market Participant of ASX Limited, a Clearing and Settlement Participant of ASX Clear Pty Limited and a Clearing Participant of ASX Clear (Futures) Pty Limited. This material is issued and distributed in Australia by or on behalf of JPMSAL only to "wholesale clients" (as defined in section 761G of the Corporations Act 2001). A list of all financial products covered can be found by visiting <https://www.jpmm.com/research/disclosures>. J.P. Morgan seeks to cover companies of relevance to the domestic and international investor base across all Global Industry Classification Standard (GICS) sectors, as well as across a range of market capitalisation sizes. If applicable, in the course of conducting public side due diligence on the subject company(ies), the Research Analyst team may at times perform such diligence through corporate engagements such as site visits, discussions with company representatives, management presentations, etc. Research issued by JPMSAL has been prepared in accordance with J.P. Morgan Australia's Research Independence Policy which can be found at the following link: [J.P. Morgan Australia - Research Independence Policy](#).

Brazil: Banco J.P. Morgan S.A. is regulated by the Comissao de Valores Mobiliarios (CVM) and by the Central Bank of Brazil. Ombudsman J.P. Morgan: 0800-7700847 / 0800-7700810 (For Hearing Impaired) / ouvidoria.jp.morgan@jpmchase.com.

Canada: J.P. Morgan Securities Canada Inc. is a registered investment dealer, regulated by the Canadian Investment Regulatory Organization and the Ontario Securities Commission and is the participating member on Canadian exchanges. This material is distributed in Canada by or on behalf of J.P.Morgan Securities Canada Inc.

Chile: Inversiones J.P. Morgan Limitada is an unregulated entity incorporated in Chile.

China: J.P. Morgan Securities (China) Company Limited has been approved by CSRC to conduct the securities investment consultancy business.

Colombia: Banco J.P. Morgan Colombia S.A. is supervised by the Superintendencia Financiera de Colombia (SFC). Any reference in this material to products or services offered abroad by entities other than the Bank in Colombia is included exclusively for descriptive purposes. Such references do not constitute, and should not be construed as, promotional activity or the provision of financial products or services within Colombian territory, as defined under applicable Colombian regulation.

Dubai International Financial Centre (DIFC): JPMorgan Chase Bank, N.A., Dubai Branch is regulated by the Dubai Financial Services Authority (DFSA) and its registered address is Dubai International Financial Centre - The Gate, West Wing, Level 3 and 9 PO Box 506551, Dubai, UAE. This material has been distributed by JP Morgan Chase Bank, N.A., Dubai Branch to persons regarded as professional clients or market counterparties as defined under the DFSA rules.

European Economic Area (EEA): Unless specified to the contrary, research is distributed in the EEA by J.P. Morgan SE ("JPM SE"), which is authorised as a credit institution by the Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and jointly supervised by the BaFin, the German Central Bank (Deutsche Bundesbank) and the European Central Bank (ECB). JPM SE is a company headquartered in Frankfurt with registered address at TaunusTurm, Taunustor 1, Frankfurt am Main, 60310, Germany. The material has been distributed in the EEA to persons regarded as professional investors (or equivalent) pursuant to Art. 4 para. 1 no. 10 and Annex II of MiFID II and its respective implementation in their home jurisdictions ("EEA professional investors"). This material must not be acted on or relied on by persons who are not EEA professional investors. Any investment or investment activity to which this material relates is only available to EEA relevant persons and will be engaged in only with EEA relevant persons.

Hong Kong: J.P. Morgan Securities (Asia Pacific) Limited (CE number AAJ321) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission in Hong Kong, and J.P. Morgan Broking (Hong Kong) Limited (CE number AAB027) is regulated by the Securities and Futures Commission in Hong Kong. JP Morgan Chase Bank, N.A., Hong Kong Branch (CE Number AAL996) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission, is organized under the laws of the United States with limited liability. Where the distribution of this material is a regulated activity in Hong Kong, the material is distributed in Hong Kong by or through J.P. Morgan Securities (Asia Pacific) Limited and/or J.P. Morgan Broking (Hong Kong) Limited.

India: J.P. Morgan India Private Limited (Corporate Identity Number - U67120MH1992FTC068724), having its registered office at J.P. Morgan Tower, Off. C.S.T. Road, Kalina, Santacruz - East, Mumbai – 400098, is registered with the Securities and Exchange Board of India (SEBI) as a 'Research Analyst' having registration number INH000001873. J.P. Morgan India Private Limited is also registered with SEBI as a member of the National Stock Exchange of India Limited and the Bombay Stock Exchange Limited (SEBI Registration Number – INZ000239730) and as a Merchant Banker (SEBI Registration Number - MB/INM000002970). Telephone: 91-22-6157 3000, Facsimile: 91-22-6157 3990 and Website: <http://www.jpimipl.com>. JPMorgan Chase Bank, N.A. - Mumbai Branch is licensed by the Reserve Bank of India (RBI) (Licence No. 53/ Licence No. BY.4/94; SEBI - IN/CUS/014/ CDSL : IN-DP-CDSL-444-2008/ IN-DP-NSDL-285-2008/ INBI00000984/ INE231311239) as a Scheduled Commercial Bank in India, which is its primary license allowing it to carry on Banking business in India and other activities, which a Bank branch in India are permitted to undertake. For non-local research material, this material is not distributed in India by J.P. Morgan India Private Limited. Compliance Officer: Ashutosh Sharma; ashutosh.j.sharma@jpmchase.com; +912261575002. Grievance Officer: Ramprasad K, jpimipl.research.feedback@jpmorgan.com; +912261573000. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Please visit [Terms and Conditions and Most Important Terms and Conditions \(MITC\)](#). The annual Compliance audit report is available at <http://www.jpimipl.com/#research>.

Indonesia: PT J.P. Morgan Sekuritas Indonesia is a member of the Indonesia Stock Exchange and is registered and supervised by the Otoritas Jasa Keuangan (OJK).

Korea: J.P. Morgan Securities (Far East) Limited, Seoul Branch, is a member of the Korea Exchange (KRX). JPMorgan Chase Bank, N.A., Seoul Branch, is licensed as a branch office of foreign bank (JPMorgan Chase Bank, N.A.) in Korea. Both entities are regulated by the Financial Services Commission (FSC) and the Financial Supervisory Service (FSS). For non-macro research material, the material is distributed in Korea by or through J.P. Morgan Securities (Far East) Limited, Seoul Branch.

Japan: JPMorgan Securities Japan Co., Ltd. and JPMorgan Chase Bank, N.A., Tokyo Branch are regulated by the Financial Services Agency in Japan.

Malaysia: This material is issued and distributed in Malaysia by JPMorgan Securities (Malaysia) Sdn Bhd (18146-X), which is a Participating Organization of Bursa Malaysia Berhad and holds a Capital Markets Services License issued by the Securities Commission in Malaysia.

Mexico: J.P. Morgan Casa de Bolsa, S.A. de C.V., J.P. Morgan Grupo Financiero is member of the Mexican Stock Exchange ("Bolsa Mexicana de Valores") and the Institutional Stock Exchange ("Bolsa Institucional de Valores"), and it is authorized to act as a broker dealer by the National Banking and Securities Exchange Commission ("Comisión Nacional Bancaria y de Valores").

New Zealand: This material is issued and distributed by JPMSAL in New Zealand only to "wholesale clients" (as defined in the Financial Markets Conduct Act 2013). JPMSAL is registered as a Financial Service Provider under the Financial Service providers (Registration and Dispute Resolution) Act of 2008.

Philippines: J.P. Morgan Securities Philippines Inc. is a Trading Participant of the Philippine Stock Exchange and a member of the Securities Clearing Corporation of the Philippines and the Securities Investor Protection Fund. It is regulated by the Securities and Exchange Commission.

Singapore: This material is issued and distributed in Singapore by or through J.P. Morgan Securities Singapore Private Limited (JPMS) [MDDI (P) 057/08/2025 and Co. Reg. No.: 199405335R], which is a member of the Singapore Exchange Securities Trading Limited, and/or JPMorgan Chase Bank, N.A., Singapore branch (JPMCB Singapore), both of which are regulated by the Monetary Authority of Singapore. This material is issued and distributed in Singapore only to accredited investors, expert investors and institutional investors, as defined in Section 4A of the Securities and Futures Act, Cap. 289 (SFA). This material is not intended to be issued or distributed to any retail investors or any other investors that do not fall into the classes of "accredited investors," "expert investors" or "institutional investors," as defined under Section 4A of the SFA. Recipients of this material in Singapore are to contact JPMS or JPMCB Singapore in respect of any matters arising from, or in connection with, the material.

South Africa: J.P. Morgan Equities South Africa Proprietary Limited and JPMorgan Chase Bank, N.A., Johannesburg Branch are members of the Johannesburg Securities Exchange and are regulated by the Financial Services Conduct Authority (FSCA).

Taiwan: J.P. Morgan Securities (Taiwan) Limited is a participant of the Taiwan Stock Exchange (company-type) and regulated by the Taiwan Securities and Futures Bureau. Material relating to equity securities is issued and distributed in Taiwan by J.P. Morgan Securities (Taiwan) Limited, subject to the license scope and the applicable laws and the regulations in Taiwan. **To the extent that J.P. Morgan Securities (Taiwan) Limited produces research materials on securities not listed on the Taiwan Stock Exchange or Taipei Exchange ("Non-Taiwan Listed Securities"), these materials shall not constitute securities recommendations for the purpose of applicable Taiwan regulations, and, for the avoidance of doubt, J.P. Morgan Securities (Taiwan) Limited does not act as broker for Non-Taiwan Listed Securities.** According to Paragraph 2, Article 7-1 of Operational Regulations Governing Securities Firms Recommending Trades in Securities to Customers (as amended or supplemented) and/or other applicable laws or regulations, please note that the recipient of this material is not permitted to engage in any activities in connection with the material that may give rise to conflicts of interests, unless otherwise disclosed in the "Important Disclosures" in this material.

Thailand: This material is issued and distributed in Thailand by JPMorgan Securities (Thailand) Ltd., which is a member of the Stock Exchange of Thailand and is regulated by the Ministry of Finance and the Securities and Exchange Commission. The registered address is 548 One City Center Building, 50th Floor, Ploenchit Road, Lumpini, Pathum Wan, Bangkok 10330.

UK: Research is produced in the UK by J.P. Morgan Securities plc ("JPMS plc") which is a member of the London Stock Exchange and is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority or J.P. Morgan Markets Limited ("JPML Ltd") which is authorised and regulated by the Financial Conduct Authority. Unless specified to the contrary, this material is distributed in the UK by JPMS plc and is directed in the UK only to: (a) persons having professional experience in matters relating to investments falling within article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) (Order) 2005 ("the FPO"); (b) persons outlined in article 49 of the FPO (high net worth companies, unincorporated associations or partnerships, the trustees of high value trusts, etc.); or (c) any persons to whom this communication may otherwise lawfully be made; all such persons being referred to as "UK relevant persons". This material must not be acted on or relied on by persons who are not UK relevant persons. Any investment or investment activity to which this material relates is only available to UK relevant persons and will be engaged in only with UK relevant persons. A description of J.P. Morgan EMEA's policy for prevention and avoidance of conflicts of interest related to the production of Research can be found at the following link: [J.P. Morgan EMEA - Research Independence Policy](#).

U.S.: J.P. Morgan Securities LLC ("JPMS") is a member of the NYSE, FINRA, SIPC, and the NFA. JPMorgan Chase Bank, N.A. is a member of the FDIC. Material published by non-U.S. affiliates is distributed in the U.S. by JPMS who accepts responsibility for its content.

General: Additional information is available upon request. The information in this material has been obtained from sources believed to be reliable. While all reasonable care has been taken to ensure that the facts stated in this material are accurate and that the forecasts, opinions and expectations contained herein are fair and reasonable, JPMorgan Chase & Co. or its affiliates and/or subsidiaries (collectively J.P. Morgan) make no representations or warranties whatsoever to the completeness or accuracy of the material provided, except with respect to any disclosures relative to J.P. Morgan and the Research Analyst's involvement with the issuer that is the subject of the material. Accordingly, no reliance should be placed on the accuracy, fairness or completeness of the information contained in this material. There may be certain discrepancies with data and/or limited content in this material as a result of calculations, adjustments, translations to different languages, and/or local regulatory restrictions, as applicable. These discrepancies should not impact the overall investment analysis, views and/or recommendations of the subject company(ies) that may be discussed in the material. Artificial intelligence tools may have been used in the preparation of this material, including assisting in data analysis, pattern recognition, and content drafting for research material. J.P. Morgan accepts no liability whatsoever for any loss arising from any use of this material or its contents, and neither J.P. Morgan nor any of its respective directors, officers or employees, shall be in any way responsible for the contents hereof, apart from the liabilities and responsibilities that may be imposed on them by the relevant regulatory authority in the jurisdiction in question, or the regulatory regime thereunder. Opinions, forecasts or projections contained in this material represent J.P. Morgan's current opinions or judgment as of the date of the material only and are therefore subject to change without notice. Periodic updates may be provided on companies/industries based on company-specific developments or announcements, market conditions or any other publicly available information. There can be no assurance that future results or events will be consistent with any such opinions, forecasts or projections, which represent only one possible outcome. Furthermore, such opinions, forecasts or projections are subject to certain risks, uncertainties and assumptions that have not been verified, and future actual results or events could differ materially. The value of, or income from, any investments referred to in this material may fluctuate and/or be affected by changes in exchange rates. All pricing is indicative as of the close of market for the securities discussed, unless otherwise stated. Past performance is not indicative of future results. Accordingly, investors may receive back less than originally invested. This material is not intended as an offer or solicitation for the purchase or sale of any financial instrument. The opinions and recommendations herein do not take into account individual client circumstances, objectives, or needs and are not intended as recommendations of particular securities, financial instruments or strategies to particular clients. This material may include views on structured securities, options, futures and other derivatives. These are complex instruments, may involve a high degree of risk and may be appropriate investments only for sophisticated investors who are capable of understanding and assuming the risks involved. The recipients of this material must make their own independent decisions regarding any securities or financial instruments mentioned herein and should seek advice from such independent financial, legal, tax or other adviser as they deem necessary. J.P. Morgan may trade as a principal on the basis of the Research Analysts' views and research, and it may also engage in transactions for its own account or for its clients' accounts in a manner inconsistent with the views taken in this material, and J.P. Morgan is under no obligation to ensure that such other communication is brought to the attention of any recipient of this material. Others within J.P. Morgan, including Strategists, Sales staff and other Research Analysts, may take views that are inconsistent with those taken in this material. Employees of J.P. Morgan not involved in the preparation of this material may have investments in the securities (or derivatives of such securities) mentioned in this material and may trade them in ways different from those discussed in this material. This material is not an advertisement for or marketing of any issuer, its products or services, or its securities in any jurisdiction.

Confidentiality and Security Notice: This transmission may contain information that is privileged, confidential, legally privileged, and/or exempt from disclosure under applicable law. If you are not the intended recipient, you are hereby notified that any disclosure, copying, distribution, or use of the information contained herein (including any reliance thereon) is STRICTLY PROHIBITED. Although this transmission and any attachments are believed to be free of any virus or other defect that might affect any computer system into which it is received and opened, it is the responsibility of the recipient to ensure that it is virus free and no responsibility is accepted by JPMorgan Chase & Co., its subsidiaries and affiliates, as applicable, for any loss or damage arising in any way from its use. If you received this transmission in error, please immediately contact the sender and destroy the material in its entirety, whether in electronic or hard copy format. This message is subject to electronic monitoring: <https://www.jpmorgan.com/disclosures/email>

MSCI: Certain information herein ("Information") is reproduced by permission of MSCI Inc., its affiliates and information providers ("MSCI") ©2026. No reproduction or dissemination of the Information is permitted without an appropriate license. MSCI MAKES NO EXPRESS OR IMPLIED WARRANTIES (INCLUDING MERCHANTABILITY OR FITNESS) AS TO THE INFORMATION AND DISCLAIMS ALL LIABILITY TO THE EXTENT PERMITTED BY LAW. No Information constitutes investment advice, except for any applicable Information from MSCI ESG Research. Subject also to [msci.com/disclaimer](https://www.msci.com/disclaimer)

Sustainalytics: Certain information, data, analyses and opinions contained herein are reproduced by permission of Sustainalytics and: (1) includes the proprietary information of Sustainalytics; (2) may not be copied or redistributed except as specifically authorized; (3) do not constitute investment advice nor an endorsement of any product or project; (4) are provided solely for informational purposes; and (5) are not warranted to be complete, accurate or timely. Sustainalytics is not responsible for any trading decisions, damages or other losses related to it or its use. The use of the data is subject to conditions available at <https://www.sustainalytics.com/legal-disclaimers>. ©2026 Sustainalytics. All Rights Reserved.

"Other Disclosures" last revised May 16, 2026.

Copyright 2026 JPMorgan Chase & Co. All rights reserved. This material or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan. It is strictly prohibited to use or share without prior written consent from J.P. Morgan any research material received from J.P. Morgan or an authorized third-party ("J.P. Morgan Data") in any third-party artificial intelligence ("AI") systems or models when such J.P. Morgan Data is accessible by a third-party.

Completed 12 Jun 2026 01:50 AM BST

Disseminated 12 Jun 2026 08:30 AM BST